MILITARY SPECIFICATION SHEET

ELECTRON TUBE, CATHODE RAY

TYPES 5UP1 AND 5UP7

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Electrostatic deflection and focus

DIMENSIONS AND PIN CONNECTIONS: See figure 1

ABSOLUTE RATINGS:

Parameter:	Ef	Ecl	ed	Eb1	Eb2	Rg	Zd	Ehk	Alt
Unit:	V	Vdc	v	Vdc	Vdc	Meg	Meg	Vdc	ft
Maximum:	6.9	-200	550	1,100	2,750	1.5	1.0	±125	10,000
Minimum:	5.7	0	---	---	1,000	---	---	---	---
TEST CONDITIONS:	6.3	Adjust	---	Focus	1,500	---	---	---	---

GENERAL:

Qualification - Required

Ⓑ denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS MINIMUM	LIMITS MAXIMUM	UNIT
	Qualification inspection						
1141	Pressure (implosion)	-		---	---	---	---
5111	Vibration	-		Width	---	1.0	mm
1331	Direct-interelectrode	-	g1 to all	Cg1	---	11	pF
			D1 to D2	C1D2	---	4	pF
			D3 to D4	C3D4	---	4	pF
			D1 to all	CD1	---	14	pF
			D2 to all	CD2	---	11	pF
			D3 to all	CD3	---	10	pF
			D4 to all	CD4	---	11	pF
5101	Neck and bulb alinement (electrostatic types)	-		Dia	---	2.25	Inch
5216	Cathode illumination	-		---	---	---	---
5246	Focusing voltage (zero-bias)	-		Eb1	260	480	Vdc
(B) 1216	Base material insulating quality	-	Zone 5 (minimum)	---	---	---	---
5248	Deflection-factor uniformity	-		---	---	---	---
	Quality conformance inspection, part 1						
5201	Electrode current (cathode)	- 2	P1; light = 4 ftL P7	Ik Ik	--- 250	1,000 ---	uAdc uAdc
5201	Voltage breakdown (electrostatic types)	-		---	---	---	---
5201	Voltage breakdown	-		---	---	---	---
5206	Gas "cross"	1 1,2	P1; light = 4 ftL P7	--- ---	--- ---	--- ---	--- ---
5106	Bulb, screen, and faceplate quality	-		---	---	---	---
5221	Light output	1	P1	Light	4.0	---	ftL
---	Screen brightness 1 (fluorescence)	1,2,3	P7	Light	0.75	---	ftL
5223	Modulation	-	P1; light = 4 ftL	ΔEc1	---	48	Vdc
5231	Spot position (electro-static deflection)	-		---	---	15	mm
5231	Spot displacement (leakage)	-		Displ	---	10	mm
5241	Grid cutoff voltage	-		Ec1	---	-67.5	Vdc
5251	Grid No. 1 leakage current	-		---	---	---	---
5251	Anode No. 2 leakage current	-		---	---	---	---
	Quality conformance inspection, part 2						
1111	Base pin solder depth	-		---	---	---	---
1301	Heater current	-		If	540	660	mA
5201	Electrode current (anode No. 1)	-	Ec1 = 0	Ib1	-15	10	uAdc

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Quality conformance inspection, part 2 -Continued						
5101	Neck and base alinement (electrostatic types)	-		---	---	---	---
5101	Angle between traces	-		---	---	---	---
5216	Stray light emission (conventional types)	-		---	---	---	---
---	Screen brightness II (phosphorescence)	2,4	P7	Light	15	---	ftL
5226	Line width A (electrostatic deflection)	- 2	P1; light = 4 ftL P7	Width Width	--- ---	0.62 0.62	mm mm
5226	Line width B (electrostatic deflection)	- 2	P1; light = 4 ftL P7	Width Width	--- ---	0.62 0.62	mm mm
---	Focusing voltage	- 2	P1; light = 4 ftL P7	Eb1 Eb1	260 260	480 480	Vdc Vdc
5248	Deflection factor	-	1D2 3D4	DF DF	42 35	58 47	Vdc/in. Vdc/in.
5251	Heater-cathode leakage current	-		---	---	---	---
(B) 5101	Base alinement (electrostatic types)	-	+3D4; pin No. 1	---	---	---	---
(B) 1101	Secureness of base, cap, or insert	-		---	---	---	---
(B) 1105	Permanence of marking	-		---	---	---	---
	Quality conformance inspection, part 3						
---	Life test	- 2	Group C; t = 500 hours; Eb2 = 2,750 Vdc; P1; light = 4 ftL P7	---	---	---	---
---	Life-test end points:						
5226	Line width A	- 2	P1; light = 3.4 ftL P7	Width Width	--- ---	0.62 0.62	mm mm
5226	Line width B	- 2	P1; light = 3.4 ftL P7	Width Width	--- ---	0.62 0.62	mm mm
5223	Modulation	-	P1; light = 3.4 ftL	ΔEc	---	48	Vdc

NOTES:

1. This test to be performed at the conclusion of the holding period.
2. Set Ec1 for ΔEc = 35 Vdc. ΔEc = grid drive from cutoff (at the voltage specified for the individual test).
3. The conditions of method 5221, light output apply except that the pattern shall be a 525-line pattern provided by 60-cycles sawtooth scanning on one of the axes.
4. Scan the tube for 10 seconds under specified conditions using a defocused 525 line 2- by 2-inch (50.80 by 50.80 mm) pattern provided by 60-cycle sawtooth scanning on one of the axes. Turn off anode voltage and read screen brightness II 15 seconds from instant anode voltages are removed.

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Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 17, 80
DSA - ES

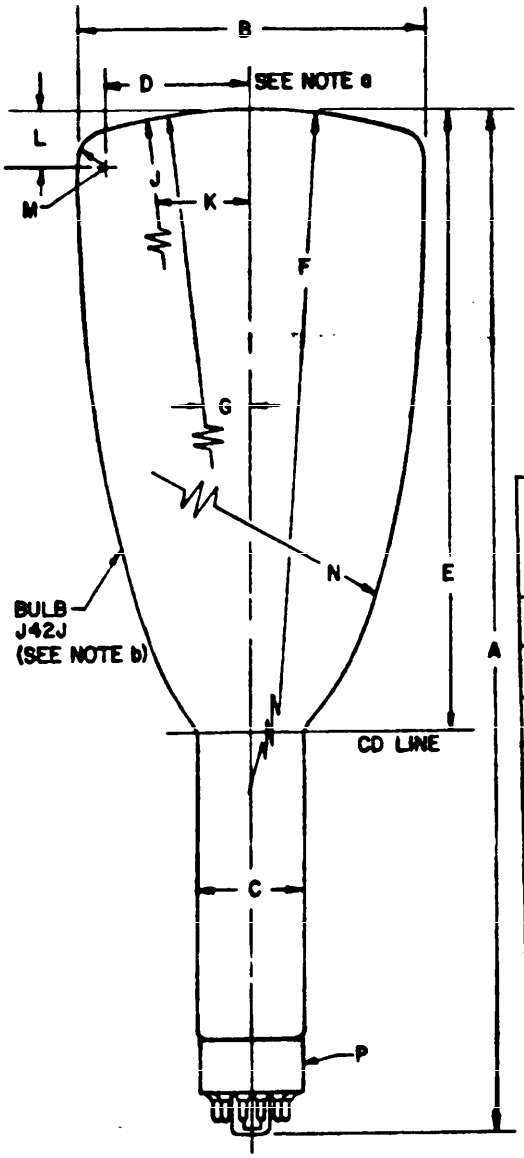
User activities:
Army - ME
Navy - AS, OS, MC, CG, SH
Air Force - 11

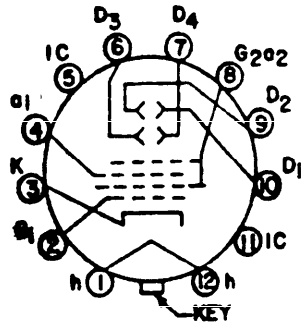
Preparing activity:
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Agent:
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BASING DIAGRAM

Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Qualification inspection		
P	Base: B12-43 or B10-75	
Quality conformance inspection, part 2		
A	14.380 (365.25)	15.130 (384.30)
B	5.156 (130.96)	5.344 (135.74)
C	1.562 (39.67)	1.688 (42.88)
D	2.250 (57.15)	---
E	---	9.380 (238.25)
Reference dimensions (see note c)		
F	20.000 (508.00) R	
G	.576 (14.63)	
H	5.360 (136.14) R	
J	1.710 (43.43) R	
K	1.444 (36.68)	
L	.808 (20.52)	
M	.500 (12.70) R	
N	20.160 (512.06) R	

NOTES:
a. Useful screen radius.
b. Clear or filter glass.
c. Reference dimensions are for information only and are not required for inspection purposes.

FIGURE 1. Outline drawing of electron tube types 5UP1 and 5UP7.

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